

Konstantinos Anagnostopoulos

Mechanical & Computer Engineer

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Started as a Mechanical Engineer with a passion for engineering and mathematics, I later discovered a strong drive for software development focused on mechanical solvers. Solving complex, real-world problems has become my core vision. With **over 8 years of experience** in automotive engineering, I specialize in the design and development of passenger and cargo vehicles. Proficient in **dynamics** and **software engineering**, my goal is to **bridge the gap between mechanical and software development**.

Knowledge

- Programming:** Applications (Desktop, Mobile, Server Web, Client Web), Embedded Systems, Machine Learning, Data Analysis, Model Based Systems Engineering, Mechanical/Mathematical Solvers
- Dynamics:** Ground Vehicles, UAVs, Multibody Dynamics, Thermal Dynamics, Control Systems, Robotics
- Structural:** FEA (Linear, Nonlinear, Explicit, Implicit), Fatigue, Modal
- Math:** ODEs, PDEs, Numerical Methods, Optimization, Frequency Domain Analysis

Job Experience

Mechanical & Software Engineer - Petros Petropoulos AEBE - Ecoshift (Athens)

Jan 2022 - now

Web/Desktop/Mobile Apps

Mechanical Design

Dynamics

FEA

Roles: Full Stack Software Developer. Dynamics Engineer. Responsible for the design and development of web, desktop and mobile applications, as well as the design and analysis of mechanical systems.

- Automotive CRM & After-Sales Management Platform**
Network Management, Lifecycle & Ticketing, DevOps & Deployment

Python

Django

PostgreSQL

REST API

DigitalOcean / CI-CD

Full Stack
- Smart Electric Scooter Mobile App (iOS & Android)**
Wireless Unlocking, Digital Key Sharing

C#

.NET MAUI

BLE

Firebase

Google Cloud
- Internal Assembly Line Production Setup Softwares**
Hardware Provisioning, UART/Modbus

C#

WPF

MAUI Blazor Hybrid

UART

Modbus

Installer
- Electric Scooter Mechanical Design**
Structural & Dynamics, Electrical Systems

SolidWorks

Autodesk Inventor

2D Drawings / GD&T

Kinematics

FEA
- Advanced Structural Dynamics & Multiphysics Optimization**
Vibration & Fatigue, Optimization Frameworks

Implicit / Explicit FEA

Topology / Shape Optimization

Multibody Dynamics (MBD)

Optimization Algorithms

Modal

Dynamics Engineer - GIVE (Athens)

Sep 2018 - Dec 2021

FEA Development

Simulation

Modeling

Mechanical Design

Dynamics

FEA

Role: Mechanical Engineer. Responsible for the development of mechanical solvers, simulation and analysis of mechanical systems, and the design of mechanical components.

- Digital Twin & Systems Engineering for Tilted-Rotor UAV - [📺 Demo](#)**
Component Selection Tool, Interactive Dashboard, Project Management

Digital Twin

Python

Plotly Dashboard

xFoil

Nonlinear ODEs

Lagrangian Mechanics

MBSE
- Electrical model of battery cell with Kalman filter for SOC estimation**
Battery modeling, Curve fitting, Kalman filter implementation

Genetic Algorithms

Nonlinear 2nd order ODEs

Kalman Filter

MATLAB/Simulink
- Leaf Spring Designer desktop app**
Finite Elements, Neural Network Integration

PyQt

Non-Linear FEA

2D Finite Elements

Newton-Raphson (quasi-static)

Classification

Neural Networks
- 1D Thermal Analysis software development - [🔗 Repository](#)**
Implementation, Testing

Python

Heat Equation

Implicit/Explicit FEA

Crank-Nicolson

Numpy

- **Electric Scooter Mechanical Design**

Dynamics, Structural

Autodesk Inventor

HyperMesh/Optistruct

MotionView/MotionSolve

Multi-Obj Optimization

Frequency Response

- **Several projects on vehicle dynamics and structural analysis of mechanical systems**

Dynamics, Structural, Excel, Hand Calculations

HyperMesh/Optistruct

MotionView/MotionSolve

Control Systems

Multi-Obj Optimization

Excel/VBA

Personal Projects

Mobile Apps

Web Development

- **Automatic save of car location for Android** - [▶ Play Store](#)

Flutter

Android Application

Bluetooth

Location Services

Background Service

- **Web Application for splitting expenses** - [🔗 Repository](#) - [🌐 Live Demo](#)

Host Server, Database, Authentication, Save as App

Python

Django

Tailscale Funnel

Ubuntu Server

SQLite

Progressive Web App (PWA)

Higher Education

Second Degree, Informatics & Telecommunication, National and Kapodistrian University of Athens

Jan 2025 - now

- **Specialization:** Computer Engineering

Integrated Bachelor & Master, Mechanical Engineering, National Technical University of Athens

Sep 2012 - Sep 2018

- **Specialization:** Ground & Aerial Vehicles
- **Grade:** 7.4/10
- **Thesis:** *Car Accident Reconstruction Using Energy Methods*
- **Published Paper:** [D. Koulocheris, C. Vossou, S. Poludoras, K. Anagnostopoulos, "The Influence of Energy Factors in the Crush Energy and DV in Vehicle Collisions", 9th International Automotive Technology Congress, OTEKON 2018](#)

Skills

Languages: Greek, English, German

Programming Languages: Python, C/C++, C#, HTML/CSS/JS, VBA, SQL & NoSQL, Fortran, Java

Frameworks & More: Django, MAUI, Flutter, Google Cloud, SKLearn, TensorFlow/Keras/Pytorch, Pandas, NumPy

Programming Software: Visual Studio, VS Code, Jupyter Notebook, Git(Hub), Linux, Docker, DBEaver

Communication: Modbus (RTU/TCP), CAN, UART, I2C, SPI, HTTP, MQTT, BLE

Mechanical Software: SolidWorks & PDM, Inventor, Altair HyperWorks, Matlab/Simulink, Webots, ROS, Gazebo

Certifications

Altair University, April 2019: HyperMesh 2017 Intermediate Examination [🔗](#)

University of Aegean, July 2019: Summer School on Industrial Product Design and Concept Development [🔗](#)

IBM/Coursera, August 2020: IBM Data Science [🔗](#)

Udemy, November 2022: Machine Learning & Self-Driving Cars: Bootcamp with Python [🔗](#)

NKUA, June 2023: Introduction to Artificial Intelligence and to Prediction & Estimation Algorithms [🔗](#)

NVIDIA, February 2026: Getting Started with Accelerated Computing in Modern CUDA C++ [🔗](#)

Other Information

Memberships

✂ [Member of the Technical Chamber of Greece \(TEE – TCG\)](#)

Personal Page

🌐 <https://ntinostempo0.github.io/Resume/>

Hobbies

Sports: Tennis, Basketball, Football, Ping Pong

Music: Piano, Clarinet, Composition

Other: Travelling, Hiking, Photography, Reading, Cooking